Vol. 137 No. 6 June 2024

Tooth numbering and classification on bitewing radiographs: an artificial intelligence pilot study

Ali Altındağ, MSc,[a] Serkan Bahrilli, MSc,[a] Özer Çelik, PhD,[b] İbrahim Şevki Bayrakdar, PhD,[c] and Kaan Orhan, PhD[d]

Objective. The aim of this study is to assess the efficacy of employing a deep learning methodology for the automated identification and enumeration of permanent teeth in bitewing radiographs. The experimental procedures and techniques employed in this study are described in the following section.

Study Design. A total of 1248 bitewing radiography images were annotated using the CranioCatch labeling program, developed in Eskişehir, Turkey. The dataset has been partitioned into 3 subsets: training (n = 1000, 80% of the total), validation (n = 124, 10% of the total), and test (n = 124, 10% of the total) sets. The images were subjected to a 3 × 3 clash operation in order to enhance the clarity of the labeled regions.

Results. The F1, sensitivity and precision results of the artificial intelligence model obtained using the Yolov5 architecture in the test dataset were found to be 0.9913, 0.9954, and 0.9873, respectively.

Conclusion. The utilization of numerical identification for teeth within deep learning-based artificial intelligence algorithms applied to bitewing radiographs has demonstrated notable efficacy. The utilization of clinical decision support system software, which is augmented by artificial intelligence, has the potential to enhance the efficiency and effectiveness of dental practitioners. (Oral Surg Oral Med Oral Pathol Oral Radiol 2024;137:679−689)

The utilization of radiological imaging techniques is of paramount importance in the comprehensive care and treatment of patients within contemporary dental practices. The utilization of radiographs, particularly periapical and bitewing films, is imperative for the visualization and analysis of dental diseases such as caries and periodontal disease, facilitating the precise identification of these conditions.[1] Dental observations and the condition of individual teeth, obtained through radiological and clinical assessments, are conventionally documented on a dental chart to ensure comprehensive patient record-keeping.[2]

Dental radiographs for clinical diagnosis can be categorized into two types: extraoral, including cone-beam computed tomography (CBCT) and panoramic images; and intraoral, comprising periapical and bitewing radiographs. CBCT provides 3-dimensional views of teeth, bone structure, and nerve pathways, while panoramic radiographs offer a comprehensive single-image view of the entire mouth. however, panoramic, periapical, and bitewing radiographs are typically 2-dimensional and are more widely accessible compared to CBCT.[3] Periapical radiographs focus on a specific area, capturing teeth from crown to root along with surrounding alveolar bone, revealing any abnormalities. Bitewing images are primarily used for caries diagnosis and bone level assessment. In the literature, investigations have been conducted pertaining to tooth numbering methodologies in panoramic, periapical, and bitewing radiographic images employing artificial intelligence algorithms.[3-8]

In response to the difficulty presented by the rigorous demands of dental procedures and in an effort to mitigate errors, there has been an increasing focus on the development of automated tooth numbering systems. The primary objective of these systems is to support healthcare professionals in optimizing their workflow efficiency and minimizing the occurrence of errors. Computer-aided systems have been developed in the domains of medical and dental imaging, thereby serving as indispensable tools for healthcare professionals.[9,10] These systems have been applied in diverse contexts, such as caries detection, staging the development of third molar teeth, identifying root frac-

[a]Department of Dentomaxillofacial Radiology, Faculty of Dentistry, Necmettin Erbakan University, Konya, Turkey.
[b]Department of Mathematics-Computer, Eskisehir Osmangazi University Faculty of Science, Eskisehir, Turkey.
[c]Department of Dentomaxillofacial Radiology, Faculty of Dentistry, Eskişehir Osmangazi University, Eskişehir, Turkey.
[d]Department of Oral and Maxillofacial Radiology, Faculty of Dentistry, Ankara University, Ankara, Turkey.
Corresponding author: Ali Altındağ. E-mail addresses: alialtindag1412@gmail.com, aaltindag@erbakan.edu.tr
Received for publication Nov 1, 2023; returned for revision Jan 13, 2024; accepted for publication Feb 8, 2024.

2212-4403/$-see front matter
https://doi.org/10.1016/j.oooo.2024.02.012

Statement of Clinical Relevance

The utilization of a YOLOv5x algorithm in conjunction with meticulously annotated training datasets within the framework of a comprehensive deep learning approach demonstrated a remarkable level of proficiency in automating tooth segmentation on dental bitewing images. This accomplishment marks a significant stride in the initial phase of diagnostic automation within this context.

tures, assessing root morphology, and detecting periodontal disease.[11-15] The application of deep learning algorithms in these contexts underscores their potential for clinical utilization.[16]

In the specialized subfield of artificial neural networks (ANNs) focused on computer vision, Convolutional Neural Networks (CNNs) are particularly prominent. CNNs employ convolution operations to create feature maps, in which each pixel or voxel's density is derived from aggregating values in the original image and the convolutional matrices, known as kernels. These kernels execute specific operations, such as sharpening or edge detection. Drawing inspiration from neurophysiological mechanisms within the brain's cortex, CNNs permit localized visual field analyses. Their layered architecture facilitates the hierarchical extraction of complex features from raw imagery, thereby optimizing feature identification.[17-20] The YOLO (You Only Look Once) algorithm, a convolutional neural network (CNN)-based deep learning model, is designed for real-time object detection. It efficiently predicts the coordinates, probability values, and object classes encapsulated within bounding boxes in images. Noteworthy for its balance of speed and accuracy, YOLO has undergone iterative development since its inception in 2016.[21,22]

The automation of tooth segmentation is a critical initial step in developing automated, interpretable diagnostic procedures for dental radiographic images. Traditional manual annotation methods are both labor-intensive and time-consuming, posing limitations for widespread application. Recent advancements in dentistry have leveraged convolutional neural networks (CNNs) to surmount the challenges inherent in traditional segmentation techniques. These modern methodologies offer a more efficient and automated approach, thereby streamlining the process of identifying tooth-related pathologies and tooth segmentation process in radiologic assessments.[18,23,24]

The objective of this study was to train a deep convolutional neural network, specifically YOLOv5, and assess its efficacy in the automated detection and numbering of teeth on bitewing radiographs.

MATERIALS AND METHODS

This study was conducted in strict compliance with ethical guidelines, having secured approval from the Ethics Committee of Necmettin Erbakan University. The authorization was conferred on April 28, 2022 under the approval number 2022/17-137. Additionally, the study adhered to the ethical principles enshrined in the Declaration of Helsinki. Checklist for Artificial Intelligence in Medical Imaging (CLAIM) and Standards for the Reporting of Diagnostic Accuracy Studies (STARD) Checklist were used for preparing the manuscript in this study.

Study sample

The study utilized archival bitewing images of individuals who sought dental treatment for various reasons at the Necmettin Erbakan University Faculty of Dentistry during the period spanning from January 2020 to September 2021. The dataset comprised a total of 1248 de-identified bitewing images.

Exclusion criteria.

- Subjects below the age of 16 years.
- Radiographic images compromised by suboptimal quality or the presence of artifacts.
- Bitewing radiographs of individuals diagnosed with specific dental pathoses, such as Dentinogenesis Imperfecta or Dentin Dysplasia.
- Bitewing images from individuals who have undergone surgical resection or oncological interventions in the maxillofacial region.

Inclusion criteria.

- Subjects aged 16 years or above.
- Presence of maxillary and mandibular posterior teeth, excluding third molars, in the radiographs.
- Radiographs devoid of extensive osseous or dental pathological manifestations.
- Subjects whose bite-wing radiographs meet diagnostic acceptability standards.

Radiographic data set

The dataset included a total of 1248 de-identified bitewing images. All bitewing radiographs were captured using the Planmeca Promax 2D (Planmeca, Helsinki, Finland) dental imaging unit with the following parameters: 68 kVp, 16 mA, 13 s.

Ground truth labeling

Ground truth annotation process was carried out using web-based annotation software (CranioCatch, Eskişehir-Turkiye) by two oral and maxillofacial radiologists (A.A. and S.B.). Before the start annotation process, a short lecture and user guide about the software and process were given to annotators about the how should be done annotation process for the well-calibration. During to annotation process, all teeth were carefully annotated from the margins by AA (10 years experience) and SB (3 years of experience) using the segmentation method using polygonal annotation tool according to FDI (Federation Dentaire Internationale) system.

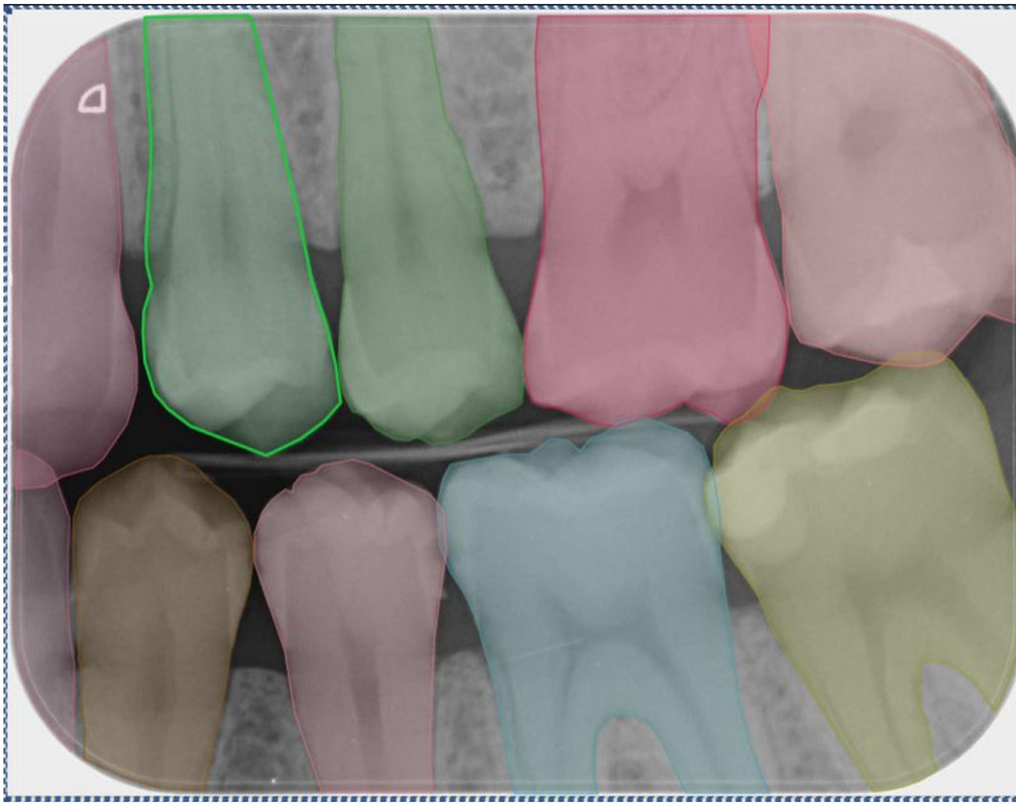

Fig. 1. Example of teeth marked from outer edges on bitewing image.

The annotation involved assigning specific numbers to each tooth, such as 16, 17, 18, 26, 27, 28, 36, 37, 38, 46, 47, and 48 for molars, and 14, 15, 24, 25, 34, 35, 44, and 45 for premolars. Additionally, the canines were labeled as 13, 23, 33, and 43. To visually represent this classification, boxes were drawn along the outer borders of the teeth (Figure 1). After the completion of annotation process, all labels were checked by annotators (A.A. and S.B.) with the exact agreement on all labels. If any conflict was seen, opinion was taken from a third oral and maxillofacial radiologist (I.S.B.- more than 10 years experience). Final decision was made with the majority voting.

Deep convolutional neural network

The basic architecture of YOLOv5 is as follows:

Backbone: YOLOv5 uses a "backbone" to perform basic feature extraction. It can be based on models such as ResNet, CSPDarknet, or EfficientNet.

Neck: This stage performs further feature extraction by combining and processing feature maps at various scales.

Head: This part is used to classify objects, determine their locations, and extract possible bounding box coordinates. Multiscale box predictions are often performed here.

Yolov5 Algorithm has 5 different models. These models are called Yolov5s, Yolov5n, Yolov5m, Yolov5l, Yolov5x. The Yolo5x model is the largest of the 5 models. Although it is slower than the others, having 88.8 million parameters increases the accuracy rate considerably. Yolov5 includes many improvements over Yolov3 and Yolov4 architectures. These improvements include techniques such as cross stage partial networks (CSP) backbone, PANet neck, Focus module, FPN module, SAM module, CBAM module, GIOU loss, CIOU loss, DIOU loss.

Yolov5x is built with a 106-layer backbone. Yolov5x's backbone is based on CSPDarknet53 architecture. Yolov5x's backbone consists of 53 ResNet blocks. Each ResNet block consists of three 3×3 convolution layers. Yolov5x's backbone contains a total of 23.8 million parameters.

Yolov5x's neck is based on PANet architecture. PANet is the combination of FPN and PAN. PANet combines feature maps of different sizes to better detect objects. The neck of Yolov5x consists of 12 PANet blocks. Each PANet block consists of two 3×3 convolution layers. Yolov5x's neck contains a total of 3.5 million parameters.

Yolov5x's head consists of 6 YOLO layers. Each YOLO layer consists of three 3×3 convolution layers. Yolov5x's head contains a total of 10.4 million parameters. The Yolov5x model uses the filters and layer numbers listed in the table below:

Layer	*Filter size*	*Filter number*
Conv1	3 × 3	128
MaxPool1	2 × 2	2
Conv2	3 × 3	256
MaxPool2	2 × 2	2
Conv3	3 × 3	512
MaxPool3	2 × 2	2
Conv4	3 × 3	1024
MaxPool4	2 × 2	2
Conv5	3 × 3	1024
PANet1	3 × 3	256
PANet2	3 × 3	512
PANet3	3 × 3	1024
YOLO5	3 × 3	64

The hyperparameters of the Yolov5x model are as follows:

- img_size= The size of the image given as input to the model. This size affects the size of objects that the model can detect.
- batch_size=Batch size is the number of data trained simultaneously from a data set. A larger batch size allows the model to learn faster.
- Classes= The number of objects that the model should detect. More classes allow the model to detect more object types.
- Learning rate = The learning rate used by the model while training. A higher learning rate allows the model to learn faster. However, a higher learning rate will result in slower training of the model.
- Optimizer: It is the optimization algorithm used by the model while training. Different optimization algorithms may cause the model to have different learning rates and performance.
- Anchor=Reference points that the model uses to detect objects. These points help the model estimate the boundaries of objects. More anchors make it easier to detect smaller objects.
- Epoch count=Epoch count is how many times a data set is trained. More number of epochs allows the model to learn better (500 used).

Hyperparameters used for augmentation;

- mosaic=1.0
- scale=0.9
- copy_paste=0.1
- hsv_s=0.7
- hsv_v=0.4
- translate=0.1

Training

The tooth detection and numbering approach proposed by the Yolov5x AI model based on a deep-Convolutional Neural Network (CNN). The bitewing images were allocated randomly among the following 3 parts:

- The training group consisted of 1000 images (8813 label),
- The validation group consisted of 124 images (1126 label),
- The testing group consisted of 124 images (1100 label).

Python, an open-source programming language (version 3.6.1; Python Software Foundation) and PyTorch library was used for developing process. The training process was handled using computer equipment of Eskisehir Osmangazi University Faculty of Dentistry Dental-AI Laboratory including Dell PowerEdge T640 Calculation Server (Dell Inc.), Dell PowerEdge T640 GPU Calculation Server (Dell Inc.), Dell PowerEdge R540 Storage Server (Dell Inc.). The training and validation datasets were utilized for the purpose of predicting and generating the most optimal weight factors for the CNN algorithm. Model hyperparameters were determined depending on the number of data, type of data, number of classes, class diversity, and success rate. Augmentation hyperparameters were used according to the amount of data for tooth segmentation in bitewing radiograph. So that the right and left side numberings do not confuse the model according to the bitewing data type, flipud and fliplr hyperparameters adjusted to flipud=0.0 and fliplr=0.0. img_size was used as 640 because resolution of the bitewing images is deemed to be sufficient. Batch size number is used as 16 taking into consideration the capacity of the graphics card and the ability to train more data at an exact time. The value of learning rate used as little as possible because of the keep success rate high. Since the number of classes was 24, the class number parameter was used as 24. SGD (Stochastic Gradient Descent) optimization algorithm, which is the main algorithm used by Yolov5, was used as the optimization algorithm. SGD is a simple and effective optimization algorithm. The basic logic of SGD is to calculate the derivative of the loss function called gradient and update the weights of the model using this gradient. According to bitewing numbering data diversity and data type, number of anchors were determined as 4. The model was trained over 500 epochs using the YOLOv5x architecture, with a learning rate of 0.01. The model is shown in Figure 2.

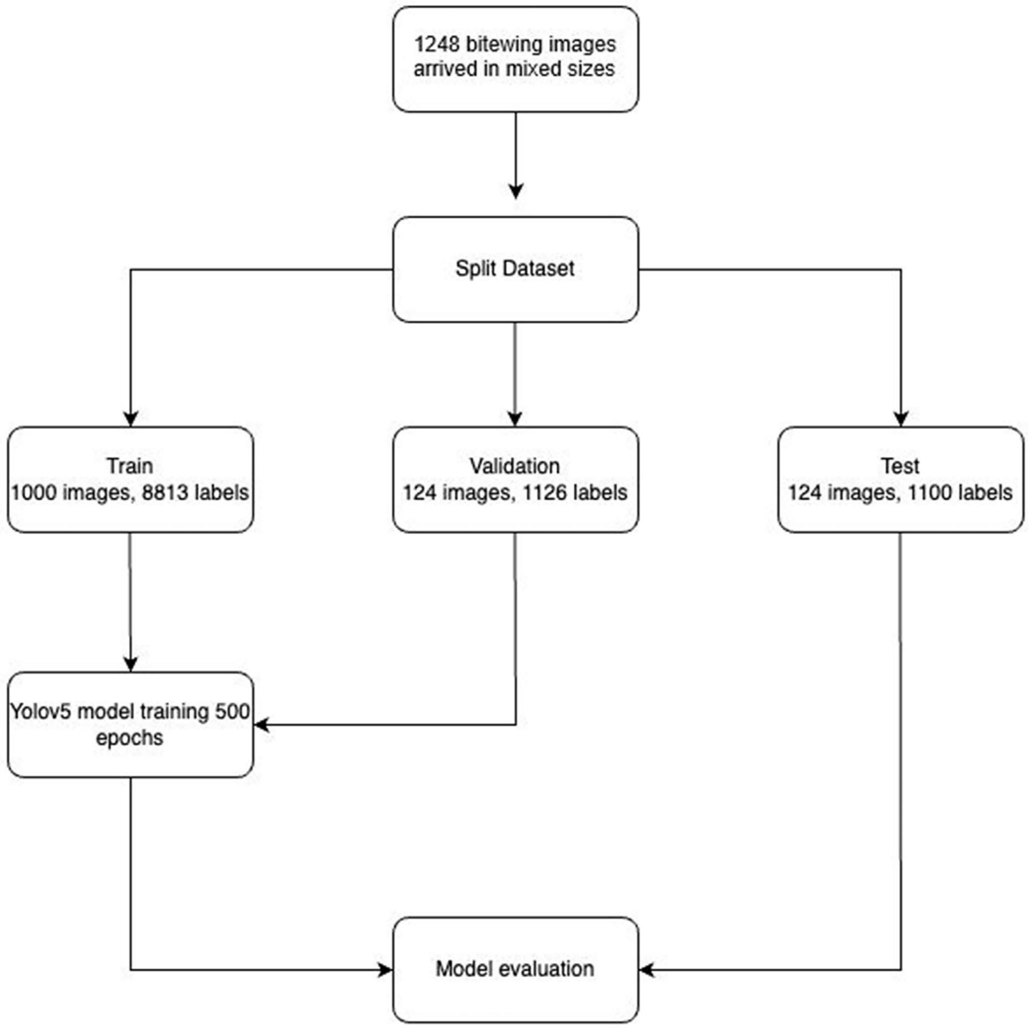

Fig. 2. The diagram of dental object detection model (CranioCatch, Eskisehir-Turkey).

Code availability

The code is publicly available at: https://github.com/ibrahimsevkibayrakdar/BitewingNumberingSegmentation

Success evaluation

The confusion matrix was employed as a performance measure to assess the efficacy of the model. The matrix presented herein serves as a comprehensive tabular representation that effectively summarizes both the anticipated and observed circumstances. The evaluation of system performance is commonly conducted through the utilization of data contained within a matrix.

The metrics employed for assessing the efficacy of the model were as follows:

A true positive (TP) refers to the accurate detection and numbering of teeth.

A false positive (FP) occurs when teeth are accurately detected but inaccurately numbered.

A false negative (FN) occurs when teeth are inaccurately identified and assigned numbers.

After performing the calculations for true positives (TP), false positives (FP), and false negatives (FN), the subsequent metrics were derived.

The sensitivity, also known as recall, is calculated by dividing the number of true positives (TP) by the sum of true positives and false negatives (TP+FN). Precision is determined by dividing the number of true positives by the sum of true positives and false positives (TP+FP). The F1 score is computed as twice the number of true positives (2TP) divided by the sum of true positives (twice), and false positives and false negatives (2TP+FP+FN).

RESULTS

Examples of correct and incorrect teeth detection boxes and teeth numbering labels were shown in Figure 3.

The utilization of deep convolutional neural network (CNN) systems in the process of numbering teeth within bitewing imaging has shown encouraging results. The utilization of advanced computational models

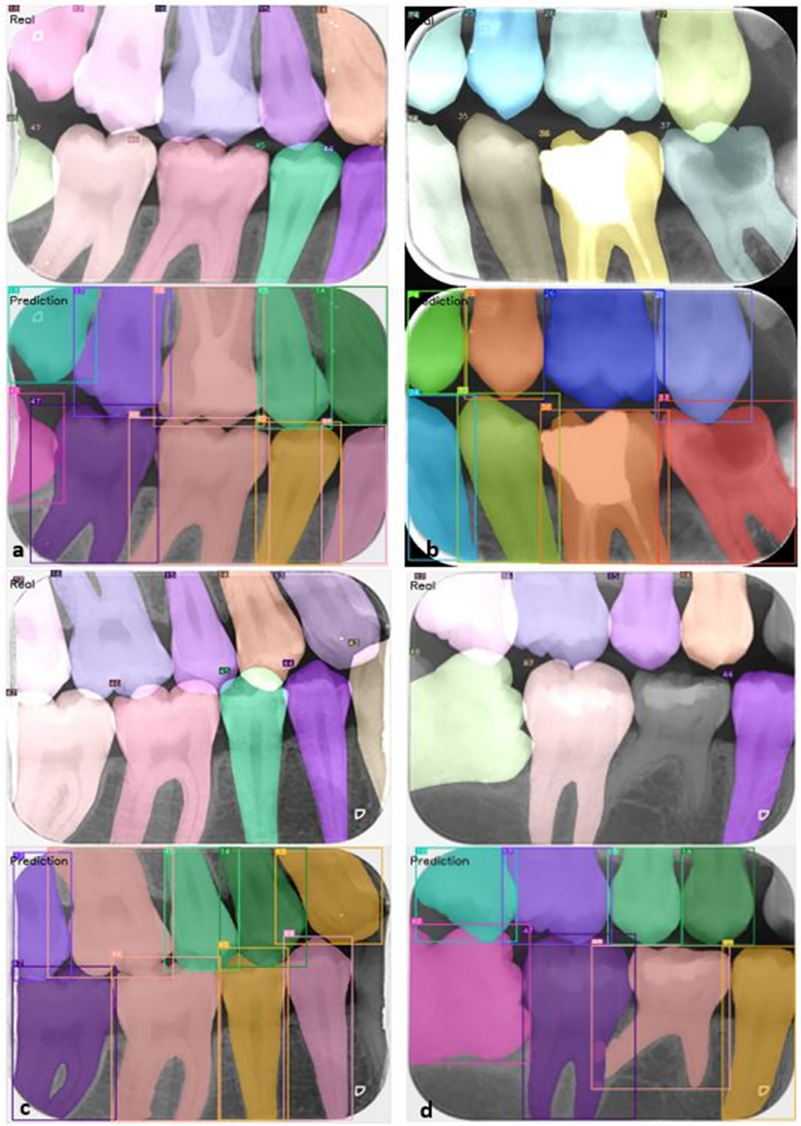

Fig. 3. Sample images correctly (A, B), and incorrectly (C, D) predicted by neural networks on test dataset.

demonstrates promising capabilities in accurately assigning tooth numbers, thereby highlighting their efficacy in optimizing dental workflows and enhancing precision in dental charting procedures. The artificial intelligence model's F1, sensitivity, and precision results, obtained through the utilization of the YOLOv5 architecture on the test dataset, were determined to be 0.9913, 0.9954, and 0.9873, correspondingly.

The training outcomes based on evaluations conducted with YOLOv5 are depicted in Figure 4. This figure schematically illustrates the fluctuations in various metrics—box loss, objectness loss, segmentation loss, classification loss, precision, recall, and mean average precision—separately for both training and validation datasets. The box loss graphs elucidate the spatial centrality of the relevant parameter, as well as its extent over the pertinent area. Objectness quantifies the likelihood of the evaluated parameter occupying the designated region, whereas the graphs for segmentation and classification losses provide insights into the predictive efficacy of the derived algorithm. To assess the model's performance, enhancements in precision, recall, and mean average precision were monitored concurrently with shifts in associated losses across 500 epochs of training. Consequently, the models exhibiting optimal performance for each parameter were identified. To facilitate comprehension, fluctuations in these metrics during the model's training phase are consolidated in Figure 4, with the peak-performing models for each parameter highlighted in red.

The performance metrics of F1-confidence curve, precision-confidence curve, precision-recall curve, and

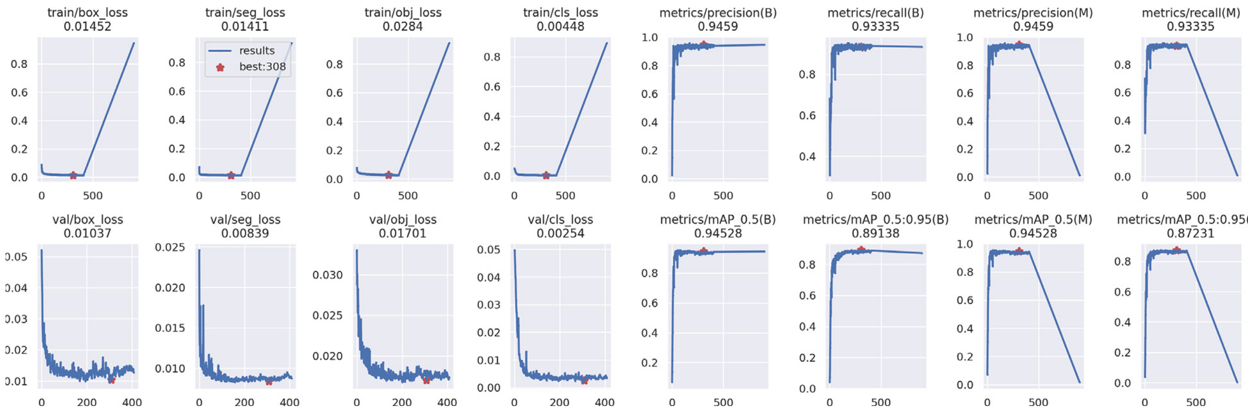

Fig. 4. The training outcomes for tooth numbering, utilizing the YOLOv5x algorithm.

recall-confidence curve were computed for each designated parameter. These metrics, which the model efficaciously analyzed, are illustrated in Figure 5. Employing correlograms, a specialized form of two-dimensional (2D) histogram, this figure facilitates the simultaneous visualization of multi-axial data, allowing for an instantaneous overview of dataset relationships and label distributions. Consequently, key attributes such as the position coordinates (x, y), width, and height of labels corresponding to pertinent parameters were integrated into a singular graphical representation. For a more nuanced visual comparison of these labels, their correlations are showcased in Figure 6.

DISCUSSION

The advent of Deep Learning and neural network methodologies has considerably accelerated the integration of artificial intelligence (AI) within the medical domain, including its burgeoning application in dentistry.[13] Numerous studies have assessed the potential accuracy of AI in interpreting medical imaging modalities such as Magnetic Resonance Imaging, radiographs, Computed Tomography, and Positron Emission Tomography, yielding promising outcomes. Although the robustness of AI models in clinical applications still necessitates comprehensive research, AI is anticipated to imminently become a fundamental component of dental diagnostics. However, there remains a need for rigorous ethical discourse surrounding its utilization.[25,26]

Machine learning (ML) and deep learning (DL) represent distinct yet interrelated computational paradigms that evolve based on experiential data. While ML demonstrates adequacy in classifying straightforward numerical data, DL excels in the analysis of expansive imaging datasets and intricate data structures.[26] Consequently, DL technologies are gaining increasing prominence within dental practice, most notably in the realm of radiology. The primary objective of AI research in dental radiology is to expedite the evaluation process for routine, uncomplicated radiographs, thereby liberating time for more complex diagnostic tasks and aiding less experienced dental practitioners in making accurate diagnoses.

Convolutional neural networks (CNNs) have been utilized in the domain of dentistry for a multitude of applications. These include the detection of caries, identification of apical pathosis, diagnosis of

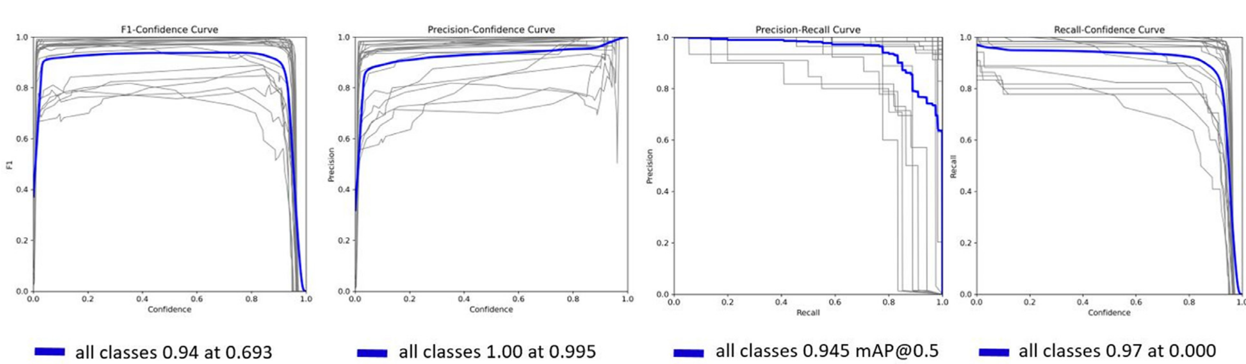

Fig. 5. The graphical representation of the e F1-confidence curve, precision-confidence curve, precision-recall curve, and recall-confidence curve.

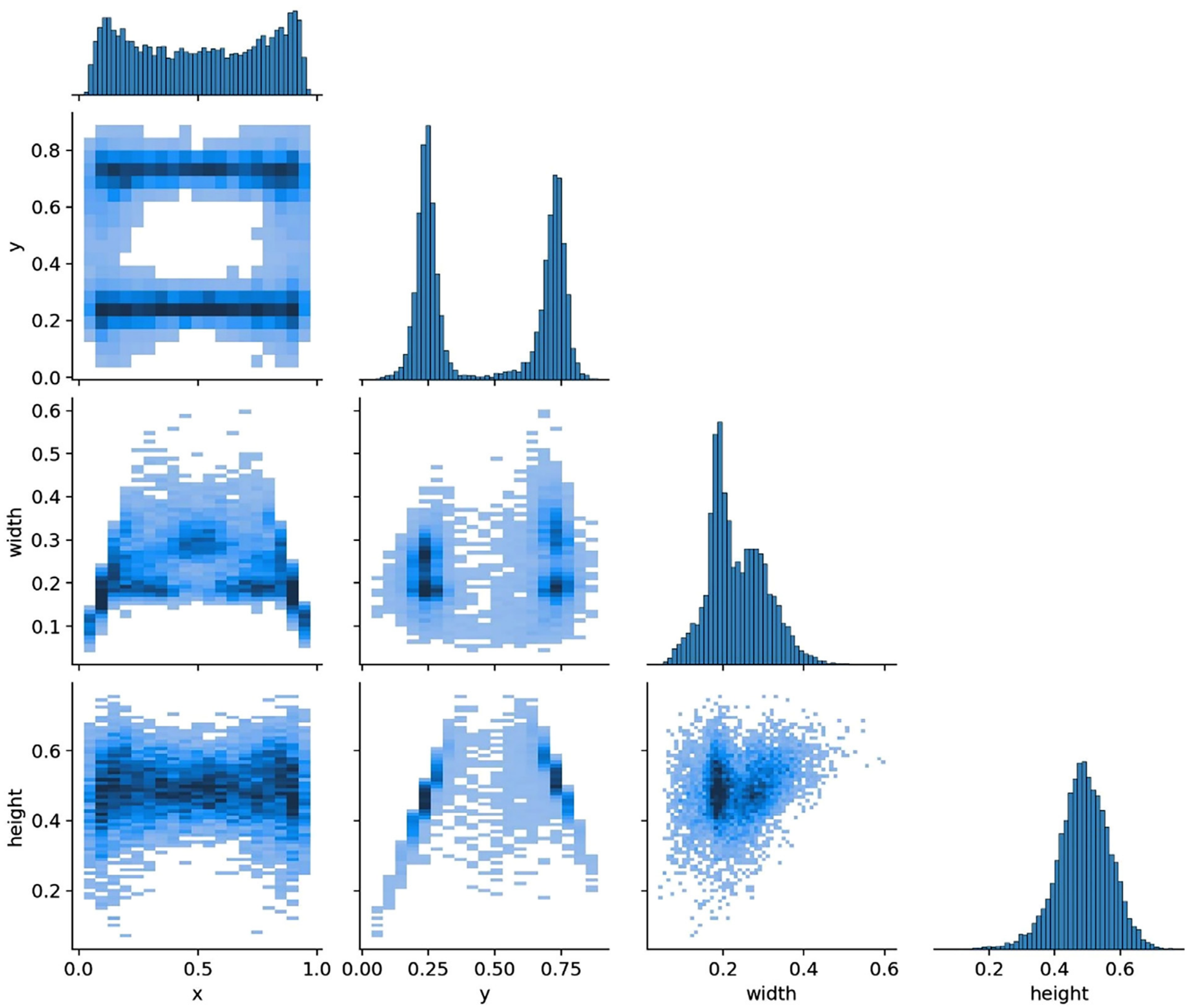

Fig. 6. Correlogram representations for dental tooth numbering labels.

periodontal diseases, detection of root fractures, diagnosis of jaw lesions, classification of maxillary sinusitis, staging of lower third molar development, localization of cephalometric landmarks, evaluation of root morphology, and detection of atherosclerotic carotid plaques.[5,14,27,28] Deep learning systems have been successfully employed in the field of dentomaxillofacial imaging, with notable applications in the classification, detection, and diagnosis of diseases, as well as the identification of anatomical structures.

Among algorithms for real-time object detection, YOLO (You Only Look Once) has emerged as a particularly favored approach. Initially introduced by Redmon in May 2016, YOLO represented a significant advancement in the field. Subsequent iterations, including YOLO 9000, YOLOv3, YOLOv4, and ultimately YOLOv5—released by Jocher in June 2020—have further refined this technique.[29] The YOLOv5 architecture is comprised of 3 primary components: the spine, neck, and head. Known for its rapid image detection capabilities, YOLOv5 exhibits superior average precision and significantly fewer background errors when compared to traditional methodologies. Furthermore, it obviates the need for complex analytical pipelines. Owing to these advantages,[30] YOLOv5 was selected as the algorithmic approach for this study.

Zhang et al.[31] conducted a study wherein deep learning techniques were employed to identify and categorize teeth in periapical radiographs. The study incorporated a dataset consisting of 1000 images, and the findings revealed notable precision and recall values of 0.958 and 0.961, correspondingly. Nevertheless, instances of misdiagnosis were observed in cases where the radiographs depicted a limited number of teeth, as the algorithm under consideration proved incapable of distinguishing between the right and left sides.

Chen et al.[11] employed the Faster R-CNN algorithm within the TensorFlow framework for the purpose of tooth detection and quantification. A total of 1250 periapical images were gathered for the purpose of evaluating the algorithm. The research emphasized the necessity of employing post-processing methods due to the initial inadequacies observed in the outcome metrics. Moreover, the authors reached the conclusion that the algorithm's performance exhibited similarity to that of a student dentist who was involved in the study.

It is imperative to acknowledge that the efficacy of deep learning algorithms based on CNNs has demonstrated significant heterogeneity in various research endeavors, tasks, and assessment criteria. The utilization of CNNs for the purpose of tooth segmentation and detection was initially investigated in the years 2015 and 2017, correspondingly. Tooth classification and detection tasks have been extensively conducted, yielding mean accuracies ranging from 0.77 to 0.98, as documented in the existing body of literature.[5]

A solitary bitewing radiograph offers a comprehensive visual representation of the occlusal surfaces of both maxillary and mandibular teeth, along with a segment of the alveolar bone. The utilization of this imaging technique is frequently employed for the detection of caries and the assessment of periodontal disease. Its effectiveness has been extensively examined in various studies documented in the literature.[32-34]

The study conducted by Lin et al.[33] employed image enhancement and tooth-segmentation methodologies to create a system with the ability to classify and assign numerical values to teeth in bitewing radiographs. The system exhibited accuracy rates of 95.1% and 98.0% for tooth classification and numbering, respectively.

The study conducted by Mahoor et al.[34] aimed to investigate dental classification in bitewing images using Bayesian classification. A notable enhancement in the performance of upper premolars was documented, as indicated by an increase in accuracy from 72.0% to 94.2%. The study's findings indicate that the classification of premolars exhibits higher efficacy in both the upper and lower jaws, in comparison to molars.

The U-Net architecture proposed by Ronneberger et al.[35] was specifically developed for the purpose of segmenting tooth structures, such as enamel, dentine, and pulp, in bitewing images. The methodology employed by the researchers resulted in a similarity score that surpassed 70%.[35,36] In a separate investigation, Eun and Kim[37] employed a CNN model to categorize tooth proposals in periapical radiographs, distinguishing them as either teeth or non-teeth. The tooth regions were delineated by the authors through the application of oriented bounding boxes. The authors' approach, which involved the utilization of oriented tooth proposals, exhibited superior performance compared to the sliding window technique. These investigations underscore the potential of utilizing advanced deep learning techniques to enhance tooth classification, numbering, and segmentation tasks in bitewing and periapical radiographic images.

Miki et al.[2] employed an automated approach for tooth classification in cone-beam computed tomography (CBCT) images, leveraging a deep CNN. They reported a charting accuracy rate of 91.0% facilitated by the deep CNN model. Consequently, they concluded that artificial intelligence (AI) presents a proficient mechanism for automating dental charting processes, and holds potential applications in the field of forensic dentistry.

In their study, Lahoud et al.[38] employed AI algorithms to facilitate automated tooth segmentation within the framework of CBCT imaging. The study yielded a mean Intersection over Union (IoU) metric of 0.87 ($\pm$0.03) when utilizing semi-automated (SA) approaches, serving as a clinical benchmark. This was juxtaposed against IoU metrics of 0.88 ($\pm$0.03) for both the fully-automated, AI-driven (F-AI) and the refined AI-driven (R-AI) methodologies. The outcomes of this study underscore the expediency and precision conferred by AI-assisted techniques in the domain of CBCT-based tooth segmentation. Notably, Lahoud et al. emphasized that such findings may herald significant advancements in the incorporation of AI-assisted modalities into the arenas of surgical planning and comprehensive oral healthcare paradigms.

Tuzof et al.[39] concentrated on the automated identification and enumeration of teeth in panoramic radiographs, utilizing a CNN-based deep learning model. Their aim was to facilitate automated dental charting in accordance with the FDI's 2-digit notation system. Specifically, they sought to number both maxillary and mandibular teeth within a single radiographic image. To achieve this, they employed the state-of-the-art Faster R-CNN model, which is underpinned by the VGG-16 convolutional architecture. Their findings were promising, suggesting that AI and deep learning algorithms hold significant promise for practical implementation within the realm of dentistry.[39]

In their study, Wirtz et al.[40] evaluated the performance of an integrated model combining a coupled shape algorithm with a neural network for the purpose of automated tooth segmentation in panoramic dental images. The model was designed to facilitate precise segmentation of the tooth region, taking into account both its spatial orientation and scale. The study yielded an average precision value of 0.790 and a recall value of 0.827, indicating promising results in the realm of automated tooth segmentation.

According to the literature, the 3 most commonly observed dental imaging modalities are periapical, bitewing, and panoramic.[5] Panoramic scans offer a comprehensive visual representation of the maxilla, mandible, and dentition. Nonetheless, their ability to accurately depict the dental roots within the osseous framework is limited. Prior research has predominantly concentrated on bitewing radiographs, which possess the inherent limitation of solely capturing the occlusal surfaces of posterior teeth. One notable constraint of this study pertains to the incomplete visualization of teeth in bitewing imaging, thereby impeding a thorough evaluation of dental conditions. When interpreting the findings presented in this study, it is crucial to take into account these limitations.

The current investigation encompasses additional constraints that warrant acknowledgment. The dataset utilized in this study comprise teeth that have undergone various dental procedures, including fillings, root canal treatments, crown-veneered restorations, as well as teeth that are in a healthy condition. Nevertheless, it is important to acknowledge that the segmentation process did not encompass deciduous teeth and teeth that were only partially visible within the imaging area. In order to improve the practicality of the study, it is recommended that future research incorporate intricate images that consider these particular scenarios.

Given these constraints, it is imperative to emphasize the significance of further endeavors and investigations prior to the operationalization of the suggested procedure as a pragmatic instrument for dental practitioners. To effectively address the disparity between research discoveries and practical clinical implementations, a more encompassing strategy is imperative. This entails the conversion of research outcomes into clinically significant effects, thereby facilitating the integration of scientific knowledge into real-world healthcare practices.

Limitations

The present study has various limitations. The dataset utilized exclusively comprised bitewing images devoid of implants, root fragments, substantial crown destruction, and deciduous teeth. Subsequent endeavors should incorporate more intricate images for a comprehensive evaluation. Another constraint involved the absence of an assessment of the experts' performance. Prior to its practical application to aid clinicians and translating the outcomes into clinically significant impacts, additional efforts and investigations are imperative for refining and advancing the established process.

CONCLUSION

The potential benefits of incorporating AI through the utilization of clinical decision support system software in dental practice are substantial, as it can significantly aid and augment the operational processes of dentists. This technology has the potential to offer significant assistance through the analysis and interpretation of intricate dental data, thereby facilitating clinical decision-making procedures.

In summary, the implementation of AI-supported clinical decision support system software exhibits considerable potential in the field of dentistry. By utilizing sophisticated algorithms and computational methodologies, these systems have the capacity to offer dentists significant assistance in the areas of diagnosis, treatment planning, and overall patient management. Further investigation and advancement in this domain are imperative in order to fully harness the capabilities of artificial intelligence technology in enhancing dental care and achieving optimal results.

FUNDING

None.

DECLARATION OF INTERESTS

There is no conflict of interest.

CREDIT AUTHOR STATEMENT

Conceptualization: AA, İŞB, KO; methodology: AA, İŞB, ÖÇ, and KO, formal analysis, ÖÇ and İŞB; investigation, AA and SB.; writig—original draft preparation, AA and SB; writing—review and editing, AA, İŞB and KO; supervision, AA, İŞB and KO. All authors have read and agreed to the published version of the manuscript.

REFERENCES

1. Vinayahalingam S, Goey RS, Kempers S, et al. Automated chart filing on panoramic radiographs using deep learning. *J Dent*. 2021;115:103864.
2. Miki Y, Muramatsu C, Hayashi T, et al. Classification of teeth in cone-beam CT using deep convolutional neural network. *Comput Biol Med*. 2017;80:24-29.
3. Kabir T, Lee CT, Chen L, Jiang X, Shams S. A comprehensive artificial intelligence framework for dental diagnosis and charting. *BMC Oral Health*. 2022;22:1-13.
4. Kılıc MC, Bayrakdar IS, Çelik Ö, et al. Artificial intelligence system for automatic deciduous tooth detection and numbering in panoramic radiographs. *Dentomaxillofac Radiol*. 2021;50(6):20200172.
5. Görürgöz C, Orhan K, Bayrakdar IŞ, et al. Performance of a convolutional neural network algorithm for tooth detection and numbering on periapical radiographs. *Dentomaxillofac Radiol*. 2022;51(3):20210246.
6. Karaoglu A, Ozcan C, Pekince A, Yasa Y. Numbering teeth in panoramic images: a novel method based on deep learning and heuristic algorithm. *Eng Sci Technol an Int*. 2023;37:101316.
7. Ari T, Sağlam H, Öksüzoğlu H, et al. Automatic feature segmentation in dental periapical radiographs. *Diagnostics*. 2022;12(12):3081.
8. Sadr S, Mohammad-Rahimi H, Ghorbanimehr MS, et al. Deep learning for tooth identification and enumeration in panoramic radiographs. *Dent Res J*. 2023;20(1):118.
9. Suzuki K. Overview of deep learning in medical imaging. *Radiol Phys Technol*. 2017;10(3):257-273.

10. Fujita H. AI-based computer-aided diagnosis (AI-CAD): the latest review to read first. *Radiol Phys Technol*. 2020;13(1):6-19.
11. Chen H, Zhang K, Lyu P, et al. A deep learning approach to automatic teeth detection and numbering based on object detection in dental periapical films. *Sci Rep*. 2019;9(1):3840.
12. Valizadeh S, Goodini M, Ehsani S, Mohseni H, Azimi F, Bakhshandeh H. Designing of a computer software for detection of approximal caries in posterior teeth. *Iran J Radiol*. 2015;12(4):e16242.
13. Orhan K, Bayrakdar IS, Ezhov M, Kravtsov A, Özyürek T. Evaluation of artificial intelligence for detecting periapical pathosis on cone-beam computed tomography scans. *Int Endod J*. 2020;53(5):680-689.
14. Ariji Y, Yanashita Y, Kutsuna S, et al. Automatic detection and classification of radiolucent lesions in the mandible on panoramic radiographs using a deep learning object detection technique. *Oral Surg Oral Med Oral Pathol Oral Radiol*. 2019;128(4):424-430.
15. Park WJ, Park JB. History and application of artificial neural networks in dentistry. *Eur J Dent*. 2018;12(04):594-601.
16. Tandon D, Rajawat J, Banerjee M. Present and future of artificial intelligence in dentistry. *J Oral Biol Craniofac Res*. 2020;10(4):391-396.
17. Hwang JJ, Jung YH, Cho BH, Heo MS. An overview of deep learning in the field of dentistry. *Imaging Sci Dent*. 2019;49(1):1-7.
18. Lee JH, Han SS, Kim YH, Lee C, Kim I. Application of a fully deep convolutional neural network to the automation of tooth segmentation on panoramic radiographs. *Oral Surg Oral Med Oral Pathol Oral Radiol*. 2020;129(6):635-642.
19. Hung K, Montalvao C, Tanaka R, Kawai T, Bornstein MM. The use and performance of artificial intelligence applications in dental and maxillofacial radiology: a systematic review. *Dentomaxillofac Radiol*. 2020;49(1):20190107.
20. Deyer T, Doshi A. Application of artificial intelligence to radiology. *Ann Transl Med*. 2019;7(11):230.
21. Bochkovskiy A, Wang CY, Liao HYM. Yolov4: optimal speed and accuracy of object detection. *arXiv preprint*. 2020:10934. arXiv.
22. Bozkaya F, Yusefi A, Tığlıoğlu Ş, et al. Otonom Sistemlerde Veri Çoğaltma Yöntemleri Kullanılarak İyileştirilmiş Gerçek Zamanlı Nesne Tespiti. *Eur J Sci Technol*. 2021;30:83-87.
23. Silva G, Oliveira L, Pithon M. Automatic segmenting teeth in X-ray images: trends, a novel data set, benchmarking and future perspectives. *Expert Syst Appl*. 2018;107:15-31.
24. Duman ŞB, Syed AZ, Celik-Ozen D, et al. Convolutional neural network performance for sella turcica segmentation and classification using CBCT images. *Diagnostics*. 2022;12(9):2244.
25. El-Damanhoury HM, Fakhruddin KS, Awad MA. Effectiveness of teaching International Caries Detection and Assessment System II and its e-learning program to freshman dental students on occlusal caries detection. *Eur J Dent*. 2014;8(04):493-497.
26. Pauwels R. A brief introduction to concepts and applications of artificial intelligence in dental imaging. *Oral Radiol*. 2021;37(1):153-160.
27. Ekert T, Krois J, Meinhold L, et al. Deep learning for the radiographic detection of apical lesions. *J Endod*. 2019;45(7):917-922.
28. Lee JH, Kim DH, Jeong SN, Choi SH. Detection and diagnosis of dental caries using a deep learning-based convolutional neural network algorithm. *J Dent*. 2018;77:106-111.
29. Rahman T, Khandakar A, Islam KR, et al. HipXNet: deep learning approaches to detect aseptic loosening of hip implants using x-ray images. *IEEE Access*. 2022;10:53359-53373.
30. Khalid M, Sarfraz MS, Iqbal U, Aftab MU, Niedbała G, Rauf HT. Real-time plant health detection using deep convolutional neural networks. *Agriculture*. 2023;13(2):510.
31. Zhang K, Wu J, Chen H, Lyu P. An effective teeth recognition method using label tree with cascade network structure. *Comput Med Imaging Graph*. 2018;68:61-70.
32. Said EH, Nassar DEM, Fahmy G, Ammar HH. Teeth segmentation in digitized dental X-ray films using mathematical morphology. *IEEE Trans Inf Forensics Secur*. 2006;1(2):178-189.
33. Lin PL, Lai YH, Huang PW. An effective classification and numbering system for dental bitewing radiographs using teeth region and contour information. *Pattern Recognit*. 2010;43(4):1380-1392.
34. Mahoor MH, Abdel-Mottaleb M. Classification and numbering of teeth in dental bitewing images. *Pattern Recognit*. 2005;38(4):577-586.
35. Ronneberger O, Fischer P, Brox T. U-net: convolutional networks for biomedical image segmentation. In: *Medical Image Computing and Computer-Assisted Intervention−MICCAI 2015: 18th International Conference*; 2015. p. Munich, Germany, October 5-9, 2015, Proceedings, Part III 18: 2015: Springer234-241.
36. Schwendicke F, Golla T, Dreher M, Krois J. Convolutional neural networks for dental image diagnostics: a scoping review. *J Dent*. 2019;91:103226.
37. Eun H, Kim C. Oriented tooth localization for periapical dental X-ray images via convolutional neural network. In: *2016 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA)*, IEEE; 2016:1-7.
38. Lahoud P, EzEldeen M, Beznik T, et al. Artificial intelligence for fast and accurate 3-dimensional tooth segmentation on cone-beam computed tomography. *J Endod*. 2021;47(5):827-835.
39. Tuzoff DV, Tuzova LN, Bornstein MM, et al. Tooth detection and numbering in panoramic radiographs using convolutional neural networks. *Dentomaxillofac Radiol*. 2019;48(4):20180051.
40. Wirtz A, Mirashi SG, Wesarg S. Automatic teeth segmentation in panoramic X-ray images using a coupled shape model in combination with a neural network. In: *Medical Image Computing and Computer Assisted Intervention−MICCAI 2018: 21st International Conference*; 2018. p. Granada, Spain, September 16-20, 2018, Proceedings, Part IV 11: 2018: Springer712-719.